
Kinetics

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Learning Objectives

After having completed this lab experiment, students will

Know:

- How the order of the reaction and the reaction rate constants as well as the reaction's activation energy can be calculated from the rate measurements
- The difference between apparent (pseudo) and true reaction rate constants

Understand:

- That the order of the reaction can only be determined experimentally by building the graphs of crystal violet (**CV**) concentrations as a function of time.

Do:

- Perform the **CV** hydrolysis reactions at different NaOH concentrations
- Observe the disappearance of the crystal violet over time as the reaction progresses using a colorimeter
- Plot [**CV**] as a function of time to determine the best equation fit and the order of the reaction.
- Determine the true reaction rate constants and the order of the reaction with respect to each reagent
- Measure the time it takes for CV to react at different temperatures and use this data to calculate the reaction's activation energy

Suggested Reading

[Chemistry: Atoms First](#) from OpenStax, Print ISBN 1947172646, Digital ISBN 1947172638, chapter 17.

Experiment Description

In this lab you will study the kinetics of the hydrolysis reaction of crystal violet with sodium hydroxide (NaOH). As the reaction proceeds, you will see the violet-colored reactant slowly change color as the colorless product forms. You will perform the reaction with different NaOH concentrations to determine the apparent (pseudo) and the true rate constants for this reaction.

Lab To Do Check List

TO DO before coming to the lab:

- Read the lab manual
- Do the pre-lab quiz to gain access to the datasheet
- Print off the data sheet and bring it with you to the lab
- Draw your procedure flowchart and bring it with you to the lab to be signed by your TA.
- Bring your student ID card, lab safety glasses and lab coat

TO DO during the lab:

- Follow your flowchart to perform the measurements of Crystal Violet hydrolysis kinetics at different NaOH concentrations and temperatures
- Fill in your Data Sheet
- Clean up

TO DO after the lab:

- Convert to electronic format and submit your signed Data Sheet as well as your signed Flowchart as a SINGLE PDF file on MyCourses by the end of your lab day.

Lab Grade Breakdown

This lab grade is 23 points total. The breakdown for this lab is as follows.

Labs	Points	Weight, % from the total lab grade	Due date
Kinetics	23	11.5%	
Pre-Lab Quiz	5		Before the start of the lab
Procedure Flowchart	5		Prepared before the start of the lab and submitted by the end of the lab day
Data Sheet/Results	10		By the end of the lab day
Preparedness & Professionalism grade	3		Assigned by your TA at the end of the lab

Background Information

Reactions can occur over a wide variety of time intervals. Some reactions are very slow. For example, it takes years for a copper roof to become green, forming basic copper carbonates and sulfates. Other reactions are very fast and almost instantaneous, such as acid-base reactions. The vast majority of reactions fall somewhere between these two extremes on the time scale. By studying the parameters affecting the rate of a reaction, chemists can find ways to speed up or slow down a reaction. The data generated during the kinetics experiments, such as reaction order, also allows us to elucidate the reaction mechanism and what the reaction pathways is. For example, if there are any intermediates, etc.

Hydrolysis of Crystal Violet

What is crystal violet?

In this experiment, you will study the reaction of crystal violet with sodium hydroxide (NaOH).

Crystal violet, also called gentian violet, is a violet dye (Figure 1) with a wide range of antibacterial, antifungal and other medicinal properties. It is also used as a dye in textile industry and for staining gram positive bacteria (Figure 2).

Have you ever had your dental hygienist stain your teeth with a violet dye to show you whether the bacterial plaques are? They probably used the crystal violet dye!



Figure 1 -

Crystal violet in solution. By LHcheM - Own work, CC BY-SA 3.0,

<https://commons.wikimedia.org/w/index.php?curid=23243595>



Figure 2 - Bacteria stained by crystal violet. By Muntasir du - Own

work, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=2699532>

Kinetics of crystal violet hydrolysis

Why are we interested in kinetics of crystal violet hydrolysis? Crystal violet is very toxic to aquatic organisms and should not be released to the waterways. Its use in textile and plastic industry produces large amounts of waste water contaminated with crystal violet (Salahudeen, N., Rasheed, A.A. Kinetics and thermodynamics of hydrolysis of crystal violet at ambient and below ambient temperatures. *Sci Rep* **10**, 21929 (2020)). This waste water can be reused in the industrial processes if the color is removed. The removal of the residual crystal violet color can be achieved by treating the water waste with sodium hydroxide to hydrolyze crystal violet and to produce colorless product (Figure 3). It is important to know the kinetics of this reaction to design optimum waste treatment procedure.

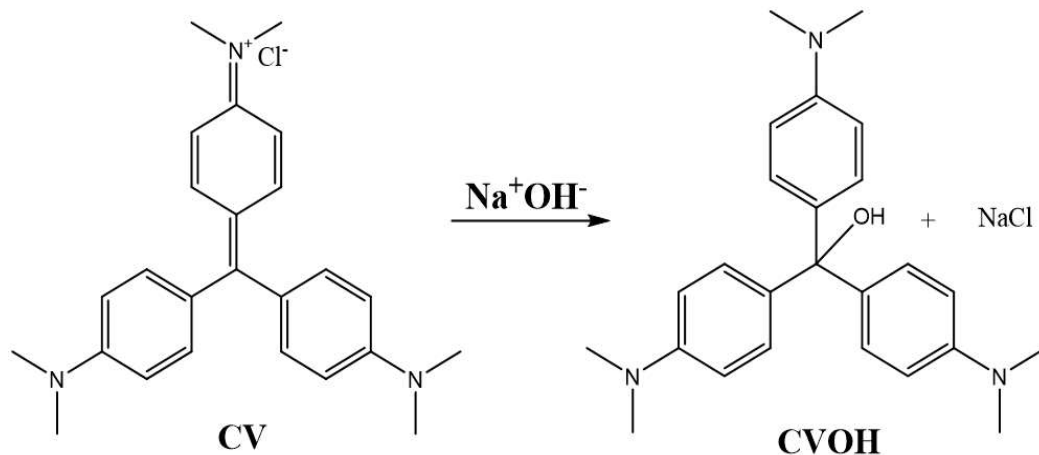
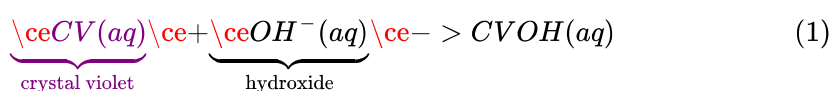


Figure 3 - Reaction of crystal violet with hydroxide ion.

We can simplify the scheme in Figure 3 to an abbreviated form:



The reaction rate and the rate law

The reaction rate of CV hydrolysis can be defined as changes in concentration of either reactants or products over time and represented by the following expression:

$$\text{reaction rate (v)} = -\frac{d[\text{CV}]}{dt} = -\frac{d[\text{OH}^-]}{dt} = \frac{d[\text{CVOH}]}{dt}$$

We can also express the reaction rate through the rate law that allows us to link the reaction rate to the concentrations of the reactants (provided

$$\text{rate} = k[\text{OH}^-]^m[\text{CV}]^n \quad (2)$$

where:

k : rate constant for the reaction

m : order of the reaction with respect to the hydroxide ion (OH^-)

n : order of the reaction with respect to crystal violet (CV)

When we do kinetics experiments, we normally want to observe a change in concentrations of the reagents over time.

As crystal violet absorbs in the visible range (maximum absorbance at 590 nm), we can monitor the changes in its concentration over time by measuring its absorbance. Absorbance is proportional to the concentration of CV (Beer's Law), and, if we know the absorption

coefficient of CV, we can later convert the absorbance to the concentration of CV. However, we cannot easily monitor changes in concentrations of both CV and NaOH simultaneously.

If we use a very large concentration of hydroxide (for example 1000 times larger than the concentration of crystal violet) then $[\text{OH}^-]$ will not change appreciably during the reaction and thus we can assign $[\text{OH}^-]^m$ as constant.

We can, therefore, re-write the equation 2 as

$$\text{rate} = k_1 [\text{CV}]^n \quad \text{eq: CVratesimple}$$

where:

$$k_1 = k [\text{OH}^-]^m \quad (4)$$

The order of the reaction and the rate constant

The order of a reaction can only be determined experimentally. Often, chemists will hypothesize that a reaction has a particular order and then test the validity of that assumption by collecting the data on how the concentrations of the reactants change over time and then performing a mathematical analysis on it.

Once the order with respect to $[\text{CV}]$ has been determined, you will also determine the rate constant (k) and order (m) with respect to NaOH for this reaction.

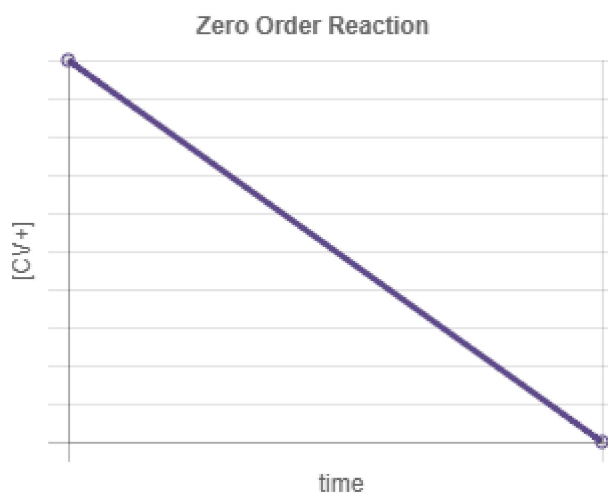
In this experiment, you will first perform the CV hydrolysis using three different concentrations of $[\text{OH}^-]$ and collect the absorbance of CV over time to test whether your data (changing concentration over time) fits best zero order, first order, or second order linear plots. The data processing will be performed during the Kinetics Data Processing lab where you will then assign the corresponding order based on which of the plots gives you the best linear fit, meaning which of the plots has the coefficient of determination R^2 closest to 1.

If the reaction is **zero order** in $[\text{CV}]$, then $n = 0$ and:

$$\text{rate} = -\frac{d[\text{CV}]}{dt} = k_1 [\text{CV}]^0 \quad \text{eq: zero_order}$$

$$\int -d[\text{CV}] = \int k_1 dt \quad (6)$$

$$[\text{CV}] = -k_1 t + [\text{CV}]_0 \quad (7)$$



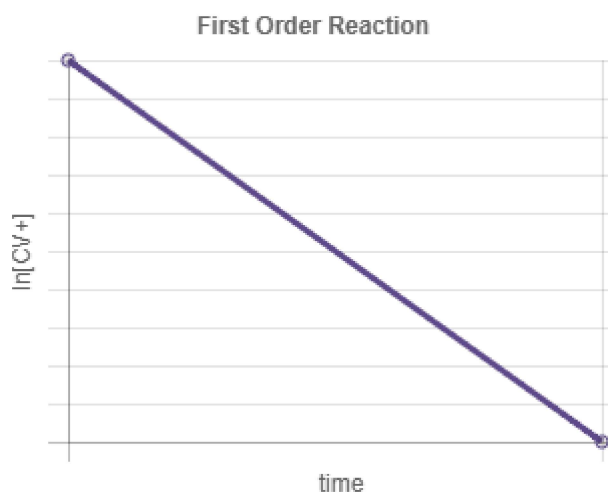
If the reaction is **first order** in CV , then $n = 1$ and:

$$\text{rate} = -\frac{d[\text{CV}]}{dt} = k_1 [\text{CV}] \quad (8)$$

$$\int \frac{d[\text{CV}]}{[\text{CV}]} = \int -k_1 dt \quad (9)$$



$$\ln[\text{CV}] = -k_1 t + \ln[\text{CV}]_0 \quad (10)$$



If the reaction is **second order** in CV , then $n = 2$ and:

$$\text{rate} = -\frac{d[\text{CV}]}{dt} = k_1 [\text{CV}]^2 \quad (11)$$

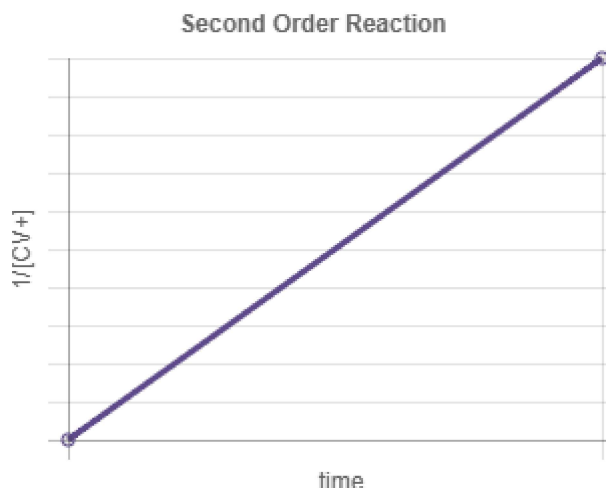


$$\int \frac{d[\text{CV}]}{[\text{CV}]^2} = \int -k_1 dt \quad (12)$$



$$\frac{1}{[\text{CV}]_t} - \frac{1}{[\text{CV}]_0} = k_1 t \quad (13)$$

$$\frac{1}{[\text{CV}]_t} = k_1 t + \frac{1}{[\text{CV}]_0} \quad (14)$$



The activation energy of CV hydrolysis

The Arrhenius equation describes the dependance of the reaction rate constant on the reaction temperature

$$k = A \times e^{-\frac{E_a}{RT}}$$

or we can re-write it in a logarithmic form as

$$\ln k = A - \frac{E_a}{RT}$$

If one can determine the reaction rate constant at different temperatures, then it is possible to calculate the activation energy of a reaction.

In this lab you will perform the CV hydrolysis at different temperatures while keeping the starting concertation of CV and NaOH constant in order to measure the reaction times and to further calculate the E_a during the Kinetics Data Processing lab.

Safety, Apparatus and Chemicals

Item	Quantity
100 mL beaker	3
25 mL graduated cylinder	1
250 mL beaker (for waste)	1
400 mL beaker (for water bath)	2
analog thermometer	2
small clamp to hold a thermometer	2
graduated pipette, 2 mL	1
graduated pipette, 5 mL	1
pipette pump	1
test tube	2
test tube rack	1
timer (cell phone can also be used)	2
Vernier interface	1
Pasteur pipette bulbs	
Pasteur pipettes	
colorimeter	1
cuvette	3
sharpie	1

Chemicals

Chemical	Quantity
0.100 M sodium hydroxide, NaOH	50 mL
5.00×10^{-4} M crystal violet aqueous solution	~12 drops in total
distilled water	

Chemical Safety and SDS Sheets



- [Crystal violet](#): corrosive and hazardous in case of ingestion or inhalation. It will also dye your hand purple for a couple of days, so you may want to wear gloves.
- [Sodium hydroxide](#): causes severe skin burns, eye damage, and respiratory irritation; corrosive

Procedure

Part 1. NaOH solution preparation

1. Label your 100 mL beakers **A**, **B** and **C**.
2. Using the 100 mL beaker labelled **A**, obtain 40 mL of the 0.100 M stock NaOH from the containers on the side bench. You will use this stock to prepare solutions **B** and **C**.
3. To prepare solution **B**, measure 20 mL of solution **A** using the 25 mL graduated cylinder and add it into the 100 mL beaker labelled **B**. Then, using the same cylinder, measure and add 20 mL of distilled water into the beaker **B**. Swirl the flask to mix the solution.
4. To prepare solution **C**, using the graduated cylinder measure and add 10 mL of solution **B** into the 100 mL beaker **C**. Then, using the same cylinder, measure and add 10 mL of distilled water into the beaker **C**. Swirl the flask to mix the solution.
5. Calculate and record the obtained concentrations of NaOH solutions on the Data Sheet using the correct number of significant figures.

Part 2. Hydrolysis of CV at different concentrations of NaOH. Data collection over time using colorimetry.

1. Prepare a blank by filling an empty cuvette 3/4 full with solution **A**. Seal the cuvette with the lid. To correctly use a colorimeter cuvette, remember:
 - All cuvettes should be wiped clean and dry on the outside with a Kimwipe tissue.
 - Handle cuvettes only by their top edge.
 - All solutions should be free of bubbles when you take absorbance readings.
 - Do not right on the sides of the cuvette. If you need to label it, please write on the cuvette lid
2. Calibrate the colorimeter as follows:
 - Holding the cuvette by its upper edges, place it in the cuvette slot of the colorimeter.
 - If the wavelength of the colorimeter isn't already set to 565 nm, press the arrow button until the 565 nm LED is lit.
 - Press the **cal** button until the LED stops blinking (around 5 seconds). You will only need to zero the instrument once; you will not need to re-zero the instrument with solutions **B** and **C**.
3. Prepare the Vernier interface for data collection by selecting **File** in the top menu, then **New**. The screen may ask you if you would like to save the current data; you should select **Discard**.



Warning

Do the following step quickly!

4. Begin the reaction by adding **2-3** drops of the crystal violet solution to the cuvette with NaOH solution **A**, cap is with a lid and, holding the lid with your index finger, turn the cuvette upside down twice. Wipe off the outside of the cuvette with a Kimwipe and gently tap the cuvette if you need to get rid of any bubbles.
5. Place the cuvette in the colorimeter and check with absorbance of the CV on the right side of the screen. If it is more than 0.7 units, wait until it drops 0.7 and then press the  at the left bottom corner of the screen. Absorbance data will be collected for 5 minutes.

6. Ensure that the run has proceeded and then discard the waste from the run into the 500 mL waste beaker. Rinse the cuvette twice with a small amount of water, but do not dry it.

7. If the absorbance level drops below 0.1 before the run is over, you can manually stop the run by pressing on the red square button at the left bottom corner of the screen.

7. The linear dynamic range (when the absorbance is directly proportional to concentration) of the colorimeters is between 0.7 and 0.1 absorbance. We do not need to have all of the data points within that range but we want as many as we can between 0.7 and 0.1. If your absorbance is above 0.8 when you insert your cuvette, wait until it drops to 0.7 and press "Start". **If the absorbance is above 0.7 for the final data point** you should **redo** the trial. You do not need change your cuvette, simply press "Start" on the Vernier apparatus and run for another 5 minutes and you should have data in the valid range.

8. **Save your run** by pressing the  icon on the right side of the screen.

Saving



The **Save** step is crucial! If you do not save your data, you will not be able to retrieve your results for your report.

8. Repeat Steps 4-7 with solutions **B** and **C**.

9. On your Data Sheet, record NaOH concentration for each run# as well the time needed for absorbance to drop from 0.7 to 0.1.

Sending your raw data

1. Send yourself a copy of the raw data by performing the following steps:
 - Tap the **File** menu at the top of the screen
 - Choose **Email** and **Text File ...**
 - In the box labelled **To**, type in your @mail.mcgill.ca e-mail address and tap on  and then . The display should inform you if the e-mail was sent correctly.
 - You should ensure the e-mail was sent to the correct address before leaving the lab. If the e-mail did not send correctly, try again or ask a TA for assistance.

- The file will compile and send all 3 sets of data that you recorded; you do not need to send multiple e-mails.

Info



The e-mail will be sent by labs.chemistry@mcgill.ca; please ensure that it does not end up in your junk mail.

Part 3. Hydrolysis of CV at different temperatures and constant concentration of NaOH

1. Choose a temperature value out of the list on the board. Write your name on the board next to your chosen value
2. Fill two 400 mL beakers with water (up to 2/3 of volume)
3. Place one of the beakers on the hotplate, insert the thermocouple probe into the beaker and set up the temperature setting on the plate according to the temperature value you have chosen.
4. Place an analog thermometer in each beaker
5. Adjust the water temperature in a second beaker such that the final temperature is anywhere between 15 - 18 °C by adding small amounts of ice
6. Using a graduated pipette, add 2.5 mL of 0.1 M NaOH in each of two test tubes
7. In each beaker, place one test tube with NaOH and leave it for 5 min to equilibrate the temperature inside the test tube to the water temperature in the beakers
8. Once the temperatures inside the test tubes have been adjusted, record the water bath temperatures on your Data Sheet
9. Add 0.200 mL of CV into each tube (one at a time) without removing it from the beaker and swirl the tube gently, while simultaneously starting the timer
10. Monitor the disappearance of color. Once you can no longer see it, press stop.
11. Record the time values on your Data Sheet

Cleanup

Dispose of the waste in your waste beaker to the appropriately labelled yellow waste container. Wash all the glassware and return to the bin.

Extra Problems (not for credit)

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